

Statement of Purpose

Applicant: Deeksha Sindhu

Department: Electrical and Computer Engineering

I still remember the first time I had my first fingerprint sensor-based smartphone. It felt astonishing how that small device could ensure the fingerprint data is validated for the same individual. How did something as soft as a fingerprint turn into a decision inside a circuit? I was probably too young to phrase it that way, but that was the moment the question slipped in: *how far can we push semiconductor physics to make electronics feel almost alive?*

That question kept me company all the way to Banasthali Vidyapith. I ended up finishing my B.Tech in ECE with a 9.43 CGPA. I am also proud to put that I hold a spot in the top ten of my department, which still feels slightly surreal when I think about the long nights that got me there. Out of everything I studied, device physics grabbed me in a way nothing else did. Those early courses in solid-state physics and digital electronics felt like someone had handed me a map, but the paths weren't drawn yet. I knew I'd have to build them myself - and the only real way to do that was through research.

My first real dive into this world happened during my major project on FeFETs at IIT Ropar under Dr. Pardeep Duhan. It was the first time I saw how theory could shift and bend when you dropped it into a simulation and expected it to behave. I started simple - PN junctions, MOS caps - and then stumbled my way toward MFS (Metal-Ferroelectric-Semiconductor) and MFIS (Metal-Ferroelectric-Insulator-Semiconductor) structures in Sentaurus TCAD.

But it was the germanium-based FeFETs that tested me. The narrow bandgap refused to "cooperate." Threshold voltage drifted. Leakage soared. Every time I thought I'd fixed one issue, another popped up like a loose wire. It sounds frustrating, and honestly, it was. But somewhere between the failed runs and the notes scribbled in margins, I realized I was learning how to sit with uncertainty. How to keep going even when a simulation refused to say yes.

And then another idea started growing in the back of my mind - almost like a whisper: what if physics alone isn't enough anymore?

That whisper turned into a full project: noise prediction in FDSOI devices. I threw every standard ML model at the dataset - ANN, RNN, SVR, Random Forest - and kept running into the same wall: 70-80% accuracy. Fine, but nowhere near satisfying.

The real shift happened when I pulled physics back into the loop and built a Physics-Informed ML model. As I anchored the Random Forest regression to its actual device behavior, I found that the predictions suddenly started making sense. The R^2 score past 90%. And that was the first time when I realised for the first time, that I wasn't choosing between physics and data science; rather I was putting them in the same room and letting them argue their way to a better answer. I'm still shaping that work into a research paper, but the core insight is already etched into me: the future isn't physics *or* AI. It's both, intertwined.

My project at IIT Jodhpur with Dr. Akshay Moudgil took me somewhere I didn't expect to go - OECTs. These things behave nothing like the silicon devices I'd grown comfortable with. Suddenly ions mattered. Electrolytes mattered. Sentaurus TCAD, which had always felt like home, turned into territory I had to stretch and reshape.

I had to “teach” a silicon-focused simulator to behave like it understood organic materials. The I-V curves I generated weren’t perfect matches with fabricated devices, but they moved in the right direction. For the first time, I could see the possibility of modeling something that wasn’t just unconventional - it was borderline unruly.

Right now, I’m building a unified current model for OECTs in Python, pulling the Bernards, Friedlein and other frameworks into one place. It fills a gap I kept running into myself: there’s no single equation that covers OECT behavior across all regimes. And with organic electronics becoming vital for wearables, neural interfaces, biomedical diagnostics - someone has to build the tools that researchers can actually rely on. I want to be that someone.

Looking back, it almost feels like each project was nudging me toward a bigger truth: traditional materials and architectures are running out of room. Scaling alone won’t save Moore’s Law. The field needs new materials - ferroelectrics, organics, hybrids. It needs architectures inspired by how brains work, not just how circuits scale. And it needs modeling techniques that understand both physics and uncertainty.

That’s the space where I want to spend my life working.

I’m drawn to emerging semiconductor devices not because they’re fashionable, but because they feel like the next chapter of a story I’ve been reading since that first fingerprint sensor moment. I want to explore devices that use strange physics - ferroelectric switching, organic charge transport, maybe even quantum effects. And I want to build modeling frameworks that don’t force us to choose between equations and data.

Beyond all the technical ambition, there’s something else that matters to me. During my Gesture Vocalizer project - built to help the deaf and dumb community communicate - I learned something I can’t forget: technology only matters when someone’s life gets easier because it exists. I don’t ever want my work to drift away from that.

What draws me to George Washington University isn’t just the reputation, though that speaks for itself. It’s the way your semiconductor research groups seem to ask the same kind of questions I’ve carried with me for years. Professor Weidong Cao’s work feels like a continuation of the path I’m already on.

And honestly, the facilities and the feeling of cross-department collaboration you’ve built there? That’s the kind of environment where I can imagine staying up late in the lab, running one more test because something in the data looks too interesting to ignore.

Every research experience I’ve had so far has taught me a different kind of patience. A different kind of stubbornness. And a different way of seeing how physics and creativity and engineering all braid together when you’re trying to build something new.

I’m ready to bring that into a PhD - not as someone looking for a degree, but as someone looking for the next place where curiosity can turn into discovery.